

DHAIRYA PATEL

Gujarat, India

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Education

VIT Bhopal University

B.Tech - Computer Science and Engineering — CGPA: 8.52

2023 – 2027

Madhya Pradesh

Experience

Research & Development Intern — [Encrypta](#)

May 2026 – Present

- Developing a Retrieval-Augmented Generation (RAG) FAQ chatbot using LangChain, ChromaDB, and Groq LLM.
- Implementing intelligent IOth clarification flow with follow-up question generation and human escalation logic.
- Building FastAPI WebSocket backend and Streamlit omnichannel UI with conversation persistence and ticketing system.

Skills

Languages: Python, SQL, C++

Data Analysis & ML: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Statistical & Time Series Analysis

Tools & Frameworks: Excel, Power BI, PostgreSQL, PySpark, AWS (Athena, Redshift), LangChain

Projects

Reinforcement Learning-Based Stock Trading Simulator | *Python, NumPy, Matplotlib*

Personal Project

- Developed Q-Learning trading agent using Geometric Brownian Motion for price simulation across 5,000+ trading episodes.
- Achieved 18% higher risk-adjusted returns (Sharpe Ratio) compared to buy-and-hold benchmark across multiple volatility regimes.
- Analyzed agent performance in high-volatility environments, validating reinforcement learning for quantitative trading strategies.

Custom Auto Sector Index Construction | *Excel, Financial Modeling*

Personal Project

- Constructed a 9-stock Auto sector index using price-weighted, market-cap, free-float, and equal-weighted methodologies.
- Performed concentration risk and volatility attribution analysis — Maruti Suzuki & M&M contributed 30–40% of total portfolio risk.
- Evaluated return distribution and risk metrics across all weighting schemes for improved sector allocation decisions.

Financial Operations Analytics System | *Python, Pandas, Statsmodels*

Personal Project

- Built revenue forecasting system using ARIMA and Prophet models targeting $\pm 5\%$ MAPE for a SaaS business.
- Developed churn prediction models with logistic regression and tree-based classifiers achieving over 80% accuracy.
- Performed CLV modeling and RFM cohort segmentation to generate insights for revenue optimization and retention strategies.

Achievements

- **ZS Associates Campus Beats Case Challenge** — Solved national-level business case on strategy, analytics, and problem-solving.
- **EY CAFTA 2026 National Case Championship** — Built treasury & hedging framework for commodity price risk and FX management.

Relevant Coursework

Mathematics & Statistics: Applied Linear Algebra, Probability and Statistics, Calculus, Numerical Methods

Computer Science: Data Structures and Algorithms, Database Management Systems, Computer Networks

Data Science & ML: Foundations of Data Science, Fundamentals of Machine Learning

Certifications

- **NISM Series VIII – Equity Derivatives** (In Progress) | SEBI Certified
- **CISCO – Networking Devices and Configurations**
- **MATLAB Onramp**